Linear Differential Equation

Linear Differential Equation: If P and Q are only functions of x or constants then the differential equation of the form $\frac{dy}{dx}+Py=Q$ is called first order linear differential equation. To solve this equation multiply it by $IF=e^{\int Pdx}$ and integrate. Solution is $y.e^{\int Pdx}=\int Q.e^{\int Pdx}+c$

$y.IF=\int Q.IF+c$

Ex: Solve $\frac{dy}{dx}+\frac{y}{x}=x^2$

Solu: We have $\frac{dy}{dx}+\frac{y}{x}=x^2$ .....(i)

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$$P=\frac{1}{x} \qquad Q=x^2 \qquad \therefore \quad IF=e^{\int Pdx}=e^{\int\frac{1}{x}dx}=e^{\ln x}=x$$

$$\therefore \quad y.x=\int x^2.xdx+c \qquad or, \quad xy=\frac{x^4}{4}+c$$

This is the required solution.

Ex: Solve $(1+x^2)\frac{dy}{dx}+2xy=4x^2$

Solu: We have $(1+x^2)\frac{dy}{dx}+2xy=4x^2$

$or, \quad \frac{dy}{dx}+\frac{2x}{1+x^2}y=\frac{4x^2}{1+x^2}$.....(i)

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$$P=\frac{2x}{1+x^2} \qquad Q=\frac{4x^2}{1+x^2} \qquad \therefore \quad IF=e^{\int\frac{2x}{1+x^2}dx}=e^{\ln(1+x^2)}=1+x^2$$

$\therefore$ solution is

$$y.(1+x^2)=\int\frac{4x^2}{1+x^2}.(1+x^2)dx+c \qquad or, \quad (1+x^2)y=\int 4x^2dx+c \qquad or, \quad (1+x^2)y=\frac{4x^3}{3}+c$$

This is the required solution.

Ex: Solve $x\frac{dy}{dx}+(x-2)y=3x^3e^{-x}$

Solu: We have

$$x\frac{dy}{dx}+(x-2)y=3x^3e^{-x} \qquad or, \quad \frac{dy}{dx}+\frac{x-2}{x}y=3x^2e^{-x}\text{.....(i)}$$

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$$P=\frac{x-2}{x} \qquad Q=3x^2e^{-x} \quad \therefore \quad IF=e^{\int\frac{x-2}{x}dx}=e^{\int\left(1-\frac{2}{x}\right)dx}=e^{\ln(x-2\ln x)}=e^{x-2\ln x}=e^xe^{-2\ln x}=e^xe^{\ln x^{-2}}=x^{-2}e^x$$

$\therefore$ solution is $\quad y.x^{-2}e^x=\int 3x^2e^{-x}x^{-2}e^x+c \quad or, \frac{ye^x}{x^2}=\int 3dx+c \quad or, \; ye^x=x^2(3x+c)$

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This is the required solution.

Ex: Solve $(1+x)\frac{dy}{dx}+3y=\frac{1+x+x^2}{(1+x)^3}$

Solu: We have

$$(1+x)\frac{dy}{dx}+3y=\frac{1+x+x^2}{(1+x)^3} \qquad or, \quad \frac{dy}{dx}+\frac{3}{1+x}y=\frac{1+x+x^2}{(1+x)^4} \quad \text{.....(i)}$$

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$$P=\frac{3}{1+x} \qquad Q=\frac{1+x+x^2}{(1+x)^4} \qquad \therefore \quad IF=e^{\int\frac{3}{1+x}dx}=e^{3\ln(1+x)}=e^{\ln(1+x)^3}=(1+x)^3$$

$\therefore$ solution is

$$y.(1+x)^3=\int\frac{1+x+x^2}{(1+x)^4}.(1+x)^3+c \qquad or,\ y.(1+x)^3=\int\frac{(1+x+x^2)}{1+x}dx+c$$

$$or,\ (1+x)^3y=\int\frac{1+x(1+x)}{1+x}dx+c \qquad or,\ (1+x)^3y=\int\left(\frac{1}{1+x}+\frac{x(1+x)}{1+x}\right)dx+c$$

$$or,\ (1+x)^3y=\int\left(\frac{1}{1+x}+x\right)dx+c \qquad or,\ (1+x)^3y=\ln(1+x)+\frac{x^2}{2}+c$$

This is the required solution.

Ex: Solve $\frac{dy}{dx}+\frac{3x^2}{1+x^3}y=\frac{\sin^2 x}{1+x^3}$

Solu: We have

$$\frac{dy}{dx}+\frac{3x^2}{1+x^3}y=\frac{\sin^2 x}{1+x^3} \quad \text{.....(i)}$$

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$$P=\frac{3x^2}{1+x^3} \qquad Q=\frac{\sin^2 x}{1+x^3} \qquad \therefore \quad IF=e^{\int\frac{3x^2}{1+x^3}dx}=e^{\ln(1+x^3)}=1+x^3$$

$\therefore$ solution is

$$y.(1+x^3)=\int\frac{\sin^2 x}{1+x^3}.(1+x^3)+c \qquad or,\ y.(1+x^3)=\int\sin^2 x dx+c$$

$$or,\ y.(1+x^3)=\frac{1}{2}\int(1-\cos 2x)dx+c \qquad or,\ (1+x^3)y=\frac{1}{2}\left(x-\frac{\sin 2x}{2}\right)+c$$

$$or,\ (1+x^3)y=\frac{x}{2}-\frac{\sin 2x}{4}+c$$

This is the required solution.

Ex: Solve $x\frac{dy}{dx}+2y=x^2\ln x$

Solu: We have

$$x\frac{dy}{dx}+2y=x^2\ln x \qquad or,\ \frac{dy}{dx}+\frac{2}{x}y=x\ln x \quad \text{.....(i)}$$

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

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$P=\frac{2}{x}$ $\quad Q=x\ln x$ $\quad \therefore \quad IF=e^{\int\frac{2}{x}dx}=e^{2\ln x}=e^{\ln x^2}=x^2$

$\therefore$ solution is

$y.x^2=\int x^3\ln x dx+c$ $\qquad$ $or,\ x^2y=\frac{1}{4}x^4\ln x-\frac{1}{4}\int x^3dx+c$

$or,\ x^2y=\frac{1}{4}x^4\ln x-\frac{1}{4}.\frac{x^4}{4}+c$ $\qquad$ $or,\ 16x^2y=4x^4\ln x-x^4+c$

This is the required solution.

Ex: Solve $(3xy-x^2)dx+x^2dy=0$

Solu: We have

$(3xy-x^2)dx+x^2dy=0$ $\quad or,\ \frac{dy}{dx}+\frac{3xy-x^2}{x^2}=0$ $\quad or,\ \frac{dy}{dx}+\frac{3}{x}y=1$

.....(i)

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$P=\frac{3}{x}$ $\quad Q=1$ $\quad \therefore \quad IF=e^{\int\frac{3}{x}dx}=e^{3\ln x}=e^{\ln x^3}=x^3$

$\therefore$ solution is

$y.x^3=\int x^3dx+c$ $\quad or,\ x^3y=\frac{x^4}{4}+c$

This is the required solution.

Ex: Solve $x(x-1)\frac{dy}{dx}-y=x^2(x-1)^2$

Solu: We have

$x(x-1)\frac{dy}{dx}-y=x^2(x-1)^2$ $\quad or,\ \frac{dy}{dx}+\frac{1}{x(x-1)}y=x(x-1)$ .....(i)

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$P=\frac{1}{x(x-1)}$ $\quad Q=x(x-1)$ $\quad \therefore \quad IF=e^{\int\frac{1}{x(x-1)}dx}=e^{\int\left(\frac{1}{x}-\frac{1}{x-1}\right)dx}=e^{\ln x-\ln(x-1)}=e^{\ln\frac{x}{x-1}}=\frac{x}{x-1}$

$\therefore$ solution is $y.\frac{x}{x-1}=\int x(x-1).\frac{x}{x-1}dx+c$ $\quad or,\ \frac{xy}{x-1}=\int x^2dx+c$ $\quad or,\ \frac{xy}{x-1}=\frac{x^3}{3}+c$

This is the required solution.

Ex: Solve $\frac{dy}{dx}+\frac{2}{x^2+1}y=\frac{1}{x^2+1}$

Solu: We have

$\frac{dy}{dx}+\frac{2}{x^2+1}y=\frac{1}{x^2+1}$ .....(i)

Comparing equ. (i) with standard equation $\frac{dy}{dx}+Py=Q$ we get,

$P=\frac{2}{x^2+1}$ $\quad Q=\frac{1}{x^2+1}$ $\quad \therefore \quad IF=e^{\int\frac{2}{x^2+1}dx}=e^{2\tan^{-1}x}$

$\therefore$ solution is

$$ye^{2\tan^{-1}x} = \int \frac{1}{x^2+1} e^{2\tan^{-1}x} dx + c$$

$$or,\ ye^{2\tan^{-1}x} = \int e^{2z} dz + c \qquad put\ \tan^{-1}x = z \qquad \therefore \frac{1}{1+x^2}dx = dz$$

$$or,\ ye^{2\tan^{-1}x} = \frac{1}{2}e^{2z} + c$$

$$or,\ ye^{2\tan^{-1}x} = \frac{1}{2}e^{2\tan^{-1}x} + c$$

$$or,\ y = \frac{1}{2} + ce^{-2\tan^{-1}x}$$

This is the required solution.

Ex: Solve $(x^3 - x)\frac{dy}{dx} - (3x^2 - 1)y = x^5 - 2x^3 + x$

Solu: We have

$$(x^3 - x)\frac{dy}{dx} - (3x^2 - 1)y = x^5 - 2x^3 + x \qquad or,\ \frac{dy}{dx} - \frac{3x^2 - 1}{x^3 - x}y = \frac{x^5 - 2x^3 + x}{(x^3 - x)}$$

$$or,\ \frac{dy}{dx} - \frac{3x^2 - 1}{x(x^2 - 1)}y = \frac{x(x^4 - 2x^2 + 1)}{x(x^2 - 1)} \qquad or,\ \frac{dy}{dx} - \frac{3x^2 - 1}{x(x^2 - 1)}y = \frac{(x^2 - 1)^2}{(x^2 - 1)}$$

$$or,\ \frac{dy}{dx} - \frac{3x^2 - 1}{x(x^2 - 1)}y = x^2 - 1 \quad \text{.....}(i)$$

Comparing equ. (i) with standard equation $\frac{dy}{dx} + Py = Q$ we get,

$$P = -\frac{3x^2 - 1}{x(x^2 - 1)} \qquad Q = x^2 - 1 \qquad \therefore \quad IF = e^{\int -\frac{3x^2-1}{x(x^2-1)}dx} = e^{-\int \frac{3x^2-1}{x^3-x}dx} = e^{-\ln(x^3-x)} = \frac{1}{x^3 - x} = \frac{1}{x(x^2 - 1)}$$

$\therefore$ solution is

$$y.\frac{1}{x(x^2 - 1)} = \int (x^2 - 1).\frac{1}{x(x^2 - 1)}dx + c \qquad or,\ \frac{y}{x(x^2 - 1)} = \int \frac{1}{x}dx + c \qquad or,\ \frac{y}{x(x^2 - 1)} = \ln x + c$$

This is the required solution.

Ex: Solve $\frac{dy}{dx} + 3x^2 y = x^5$

Solu: We have

$$\frac{dy}{dx} + 3x^2 y = x^5 \text{.....:(i)}$$

Comparing equ. (i) with standard equation $\frac{dy}{dx} + Py = Q$ we get,

$$P = 3x^2 \qquad Q = x^5 \qquad \therefore \quad IF = e^{\int 3x^2 dx} = e^{x^3}$$

$\therefore$ solution is

$$y.e^{x^3} = \int x^5.e^{x^3}dx + c$$

$$or,\ ye^{x^3} = \int x^2.x^3.e^{x^3}dx + c \qquad put \quad x^3 = z \qquad \therefore\ 3x^2dx = dz$$

$$or,\ ye^{x^3} = \frac{1}{3}\int ze^z dz + c$$

$$or,\ ye^{x^3} = \frac{1}{3}\left[ze^z - \int e^z dz\right] + c \qquad or,\ ye^{x^3} = \frac{1}{3}\left[ze^z - e^z\right] + c$$

$$or,\ ye^{x^3} = \frac{1}{3}\left[x^3e^{x^3} - e^{x^3}\right] + c \qquad or,\ y = \frac{1}{3}(x^3 - 1) + ce^{-x^3}$$

This is the required solution.

Bernoulli's Equation:

If P and Q are only functions of x or constants then the differential equation of the form $\frac{dy}{dx} + Py = Qy^n$; $n \neq 1$ is called Bernoulli's equation. Dividing this equation by y^n it reduces to linear differential equation.

Ex: Solve $\frac{dy}{dx} + \frac{2y}{x} = \frac{y^3}{x^3}$

Solu: We have $\frac{dy}{dx} + \frac{2y}{x} = \frac{y^3}{x^3}$

This is a Bernoulli's equation

Dividing both sides by y^3 we get

$$\frac{1}{y^3}\frac{dy}{dx} + \frac{2}{xy^2} = \frac{1}{x^3} \text{.....(i)}$$

Putting $\frac{1}{y^2} = z \qquad -\frac{2}{y^3}\frac{dy}{dx} = \frac{dz}{dx} \qquad \therefore \frac{1}{y^3}\frac{dy}{dx} = -\frac{1}{2}\frac{dz}{dx}$

Equation (i) becomes

$$-\frac{1}{2}\frac{dz}{dx} + \frac{2z}{x} = \frac{1}{x^3} \qquad or,\ \frac{dz}{dx} - \frac{4z}{x} = \frac{-2}{x^3}$$

This is a linear equation with dependent variable z

$$Here,\quad P = \frac{-4}{x} \qquad Q = \frac{-2}{x^3} \qquad \therefore \quad IF = e^{\int \frac{-4}{x}dx} = e^{-4\ln x} = e^{\ln x^{-4}} = \frac{1}{x^4}$$

$\therefore$ solution is

$$z.\frac{1}{x^4} = \int \frac{-2}{x^3}.\frac{1}{x^4}dx \qquad or,\ \frac{z}{x^4} = -2\int \frac{1}{x^7}dx + c$$

$$or,\ \frac{z}{x^4} = -2.\frac{x^{-6}}{-6} + c \qquad or,\ \frac{1}{x^4y^2} = \frac{1}{3x^6} + c$$

This is the required solution.

Ex: Solve $y(2xy + e^x)dx - e^x dy = 0$

Solu: We have

$y(2xy+e^x)dx - e^x dy = 0$ or, $y(2xy+e^x)dx = e^x dy$

or, $\frac{dy}{dx} = \frac{y(2xy+e^x)}{e^x}$ or, $\frac{dy}{dx} = \frac{2xy^2+ye^x}{e^x} = 2xy^2e^{-x} + y$

or, $\frac{dy}{dx} - y = 2xy^2e^{-x}$

This is a Bernoulli's equation

Dividing both sides by y^2 we get

$\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y} = 2xe^{-x}$ .....(i)

Putting $\frac{1}{y} = z$ $-\frac{1}{y^2}\frac{dy}{dx} = \frac{dz}{dx}$ $\therefore \frac{1}{y^2}\frac{dy}{dx} = -\frac{dz}{dx}$

Equation (i) becomes

$-\frac{dz}{dx} - z = 2xe^{-x}$ or, $\frac{dz}{dx} + z = -2xe^{-x}$

This is a linear equation with dependent variable z

Here, $P = 1$ $Q = -2xe^{-x}$ $\therefore$ $IF = e^{\int dx} = e^x$

$\therefore$ solution is

$z.e^x = \int -2xe^{-x}e^x dx$ or, $ze^x = -2\int xdx + c$ or, $\frac{e^x}{y} = -x^2 + c$ or, $e^x = y(-x^2+c)$

This is the required solution.

Ex: Solve $(x-y^2)dx + 2xydy = 0$

Solu: We have

$(x-y^2)dx + 2xydy = 0$ or, $(x-y^2)dx = -2xydy$ or, $\frac{dy}{dx} = \frac{x-y^2}{-2xy}$

or, $\frac{dy}{dx} = -\frac{1}{2y} + \frac{y}{2x}$ or, $\frac{dy}{dx} - \frac{y}{2x} = -\frac{1}{2}y^{-1}$

This is a Bernoulli's equation

Dividing both sides by y^{-1} we get

$\frac{1}{y^{-1}}\frac{dy}{dx} - \frac{1}{2y^{-2}x} = -\frac{1}{2}$ .....(i)

Putting $\frac{1}{y^{-2}} = z$ or, $y^2 = z$ or, $2y\frac{dy}{dx} = \frac{dz}{dx}$ $\therefore$ $\frac{1}{y^{-1}}\frac{dy}{dx} = \frac{1}{2}\frac{dz}{dx}$

Equation (i) becomes

$\frac{1}{2}\frac{dz}{dx} - \frac{z}{2x} = -\frac{1}{2}$ or, $\frac{dz}{dx} - \frac{z}{x} = -1$

This is a linear equation with dependent variable z

Here, $P = 1$ $Q = -2xe^{-x}$ $\therefore$ $IF = e^{\int dx} = e^x$

$\therefore$ solution is

$z.e^x = \int -2xe^{-x}e^x dx$ or, $ze^x = -2\int xdx + c$ or, $\frac{e^x}{y} = -x^2 + c$ or, $e^x = y(-x^2+c)$

This is the required solution.

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$\frac{dy}{dx} - xy =$

Ex: Solve $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

Solu: We have

$$xy - \frac{dy}{dx} = y^3 e^{-x^2} \qquad or, \quad \frac{dy}{dx} - xy = -y^3 e^{-x^2}$$

This is a Bernoulli's equation

Dividing both sides by y^3 we get

$$\frac{1}{y^3}\frac{dy}{dx} - \frac{x}{y^2} = -e^{-x^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\text{(i)}$$

Putting $\frac{1}{y^2} = z \qquad \therefore \quad \frac{-2}{y^3}\frac{dy}{dx} = \frac{dz}{dx} \qquad or, \quad \frac{1}{y^3}\frac{dy}{dx} = -\frac{1}{2}\frac{dz}{dx}$

Equation (i) becomes

$$-\frac{1}{2}\frac{dz}{dx} - zx = -e^{-x^2} \qquad or, \frac{dz}{dx} + 2zx = 2e^{-x^2}$$

This is a linear equation with dependent variable z

Here, $P = 2x \qquad Q = 2e^{-x^2} \qquad \therefore \quad IF = e^{\int 2x dx} = e^{x^2}$

$\therefore$ solution is

$$ze^{x^2} = \int 2e^{-x^2} e^{x^2} dx + c \qquad or, ze^{x^2} = 2\int dx + c \qquad or, \frac{e^{x^2}}{y^2} = 2x + c \qquad or, \; e^{x^2} = y^2(2x+c)$$

This is the required solution.

Ex: Solve $\frac{dy}{dx} - 2y\tan x = y^2 \tan^2 x$

Solu: We have $\frac{dy}{dx} - 2y\tan x = y^2 \tan^2 x$

This is a Bernoulli's equation

Dividing both sides by y^2 we get

$$\frac{1}{y^2}\frac{dy}{dx} - \frac{2\tan x}{y} = \tan^2 x \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\text{(i)}$$

Putting $\frac{1}{y} = z \qquad \therefore \quad \frac{-1}{y^2}\frac{dy}{dx} = \frac{dz}{dx} \qquad or, \quad \frac{1}{y^2}\frac{dy}{dx} = -\frac{dz}{dx}$

Equation (i) becomes

$$-\frac{dz}{dx} - 2z\tan x = \tan^2 x \qquad or, \frac{dz}{dx} + 2z\tan x = -\tan^2 x$$

This is a linear equation with dependent variable z

Here, $P = 2\tan x \qquad Q = -\tan^2 x$

$\therefore \quad IF = e^{\int 2\tan x dx} = e^{2\ln(\sec x)} = \sec^2 x$

$\therefore$ solution is

$$z\sec^2 x = -\int \tan^2 x se^2 x dx + c$$

$$or, z\sec^2 x = -\int \tan^2 x \sec^2 x dx + c \qquad put \quad \tan x = u \qquad \therefore \sec^2 x dx = du$$

$$or, z\sec^2 x = -\int u^2 du + c$$

$$or, z\sec^2 x = -\frac{u^3}{3} + c \qquad or, \quad \frac{\sec^2 x}{y} = -\frac{1}{3}\tan^3 x + c$$

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